# Indian Institute of Science Education and Research Kolkata

## CH2203

## Synthesis and Characterization Laboratory
## (Organic Chemistry)

## Laboratory Manual

IISER KOLKATA

## Spring Semester 2020

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**Experiment 1: Purification of Benzoic Acid by Recrystallization**

**Purpose:** To purify samples of organic compounds that are solids at room temperature.

The products of chemical reactions can be impure. Purification of your products must be performed to remove by-products and impurities. Liquids are customarily purified by distillation, while solids are purified by recrystallization (sometimes called simply "**crystallization**").

When organic substances are synthesized in the laboratory or isolated from plants, they contain impurities. Several techniques for purifying these compounds have been developed. The most basic of these techniques for the purification of organic solids is recrystallization, which relies on the different solubilities of solutes in a solvent. Less soluble compounds will crystallize first. The crystallization process helps in the purification because as the crystals form, they select the correct molecules that fit into the crystal lattice and ignore the wrong molecules. This is, of course, a process that could be improved, but it does increase the purity of the final product.

The solubility of the compound in the solvent used for recrystallization is essential. In the ideal case, the solvent completely dissolves the compound to be purified at a high temperature. Usually, the boiling point of the solvent and the compound would be utterly insoluble in that solvent at room temperature or 0 °C. In addition, the impurity would either be completely insoluble in the particular solvent at a high temperature or very soluble at a low temperature. In the former case, the impurity could be filtered off at high temperatures, while in the latter case, the impurity would stay entirely in solution upon cooling. This will never happen in the real world, and recrystallization is a technique that must be practiced and perfected.

Regardless of the crystallization method, the purity of the solid can be verified by taking the melting point. A good (suitable) recrystallization solvent will dissolve a large amount of the impure compound at temperatures near the boiling point of the solvent. A small amount of the compound being purified should remain in solution at low temperatures, between approximately 25 and –5 °C. Low solubility at low temperatures minimizes the amount of purified compound lost during recrystallization.

A suitable recrystallization solvent should also be partially volatile to remove from the purified crystals easily. The solvent should not react with the compound being purified, and it should have a boiling point below the melting point of the compound being purified because the solid melts before it dissolves (oiling out). When selecting a suitable recrystallization solvent, one should consider flammability, toxicity, and expense.

In selecting a solvent, consider that like likes like. Polar compounds dissolve polar compounds, and non-polar compounds dissolve non-polar compounds. The most commonly used recrystallization solvents are presented in the following table.

| Solvent | Polarity | Boiling point (°C) |
|---|---|---|
| water | very polar | 100 |
| ethanol | polar | 78 |
| methanol | polar | 65 |
| dichloromethane | slightly polar | 40 |
| diethyl ether | slightly polar | 35 |

Organic compounds with one polar functional group and low carbon atoms such as methanol, ethanol, and n-propanol are highly soluble (miscible) in water. These alcohols form hydrogen bonds with water due to the polar –OH functional group. As the number of carbons per polar functional group increases, solubility decreases. The solubility of alcohols with four to five carbons is given in the following table.

**Experimental Procedures:**

Using a weighing paper, weigh about 1.00 g of "impure Benzoic acid for recrystallization" and transfer it to a 125 mL Erlenmeyer flask. Add 20 mL of distilled water to the flask using a graduated cylinder. Bring the mixture to the boiling point by heating it on a hot plate, stirring the mixture, and boiling gently to dissolve benzoic acid completely.

| alcohol | Solubility (g/100 mL $H_2O$) |
|---|---|
| n-butanol | 8 |
| n-pentanol | 2 |
| n-hexanol | 0.5 |
| n-heptanol | 0.1 |

Remove the flask from the hot plate and examine the solution. If particles of benzoic acid are still undissolved, add an additional amount of hot or cold water in small increments and resume heating the solution. The objective is to dissolve the entire solid in as much hot or near-boiling solvent (water) as necessary. Do not add too much water; the solution will not be saturated, and the yield of purified benzoic acid will be reduced. Add water in small amounts (several drops from a Pasteur pipette) until all the benzoic acid is dissolved and the solution is boiling.

If the solution is completely clear (though not necessarily colorless) and no solid benzoic acid is visible, then add 10-15 ml water to the mixture, place the Erlenmeyer flask on a countertop where it will not be disturbed, and cover it with an upside-down small beaker (to prevent dust contamination). Allowing the flask to cool slowly will give the best-shaped crystals after about 5-10 minutes. If crystallization does not occur after 10 minutes, scrape the sides of the flask above the level of the solution with the sharp end of a glass rod hard enough to audibly scratch the interior surface of the flask. This may dislodge some undetectable, tiny crystals that will drop into the solution and "seed" the solution, helping to induce crystallization. A seed crystal can serve as a nucleation point for the crystallization process. Cooling the solution in an ice bath may also help at this point.

When the crystals have formed completely (may require an ice bath), collect your solid chemical by setting up a vacuum (suction) filtration on a properly fitted filter paper in a clean Büchner funnel apparatus. Pour the chilled mixture into the Buchner funnel. The water should filter quickly. Get all the crystals out of the flask using a spatula or stirring rod. Rinsing with 1 or 2 mL of cold water helps remove the crystals from the flask, and rinsing helps remove impurities.

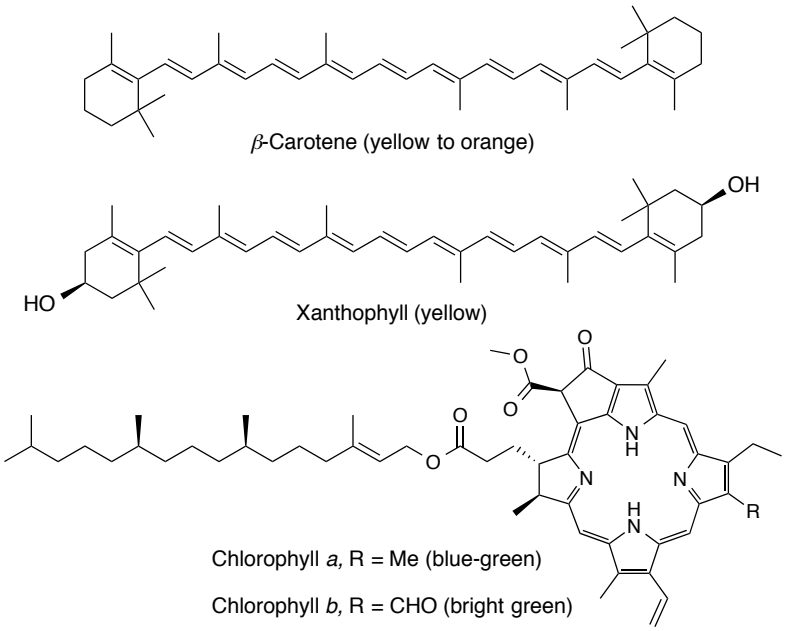
Let the aspirator run for a few minutes to start air-drying the crystals. Then, use a spatula to lift the filter paper and crystals out of the Buchner funnel, press them as dry as possible on a large clean paper towel (hand dry), allow them to dry completely, and transfer the dry sample to a pre-weighing paper. Determine the weight of the DRY crystals of recovered benzoic acid.

Calculate the percent recovered using the following *written* formula and determine the melting point of your recrystallized benzoic acid.

$$\textbf{\% Recovered} = \frac{\textbf{Weight of benzoic acid obtained after recrystallization}}{\textbf{Weight of benzoic acid before recrystallization}} \textbf{x100}$$

## Experiment 2: Separation of Spinach Pigments by Thin Layer Chromatography (TLC)

Chromatography has been a fundamental technique used for chemical separation that dates back to the 1850's. The current experiment emphasizes column and thin-layer chromatography for isolating and identifying photosynthetic pigments from spinach leaves. The leaves of plants contain, in addition to green chlorophyll *a* and *b*, other pigments that become visible in the fall as leaves die and the chlorophyll rapidly decomposes. The carotenoids are among the most abundant of these other pigments, including beta-carotene, xanthophylls, and chlorophyll *a* and *b*. Carotenoids provide most yellow, orange, and red colors in fruits and vegetables. In this experiment, solvent extraction followed by column chromatography will be used to isolate *β*-Carotene, Xanthophylls, and Chlorophyll *a* from spinach leaves.




**Experimental Procedure:**

**Pigment Extraction:**

1. On balance, weigh 0.5 g of fresh spinach and combine it with 1.5 grams of anhydrous magnesium sulfate (or sodium sulfate). Transfer these materials to a mortar and, using a pestle, grind the mixture until a fine dry powder is obtained. The anhydrous magnesium sulfate (or sodium sulfate) will remove the water from the leaves.

2. Transfer the powder (2.0 g total) to a small test tube and combine with 2.0 mL of acetone. Shake the test tube vigorously for approximately one minute. You need to make sure that the solid and solvent are well mixed.

3. Allow this mixture to stand for 10 minutes. Then, carefully transfer the solvent above the solid using a pipet into a small tube. Do not transfer any of the solid material. The solvent extract should be green in color. Cap the tube to minimize solvent evaporation.

4. Obtain a TLC chamber (a glass jar with a cover or beaker covered with a watch glass or aluminium foil) and add developing solvent (a mixture of petroleum ether, acetone, ethyl acetate and methanol). The solvent should completely cover the bottom of the chamber to a depth of approximately 0.5 cm.

**Performing TLC**

1. Obtain a TLC plate that has been precut (1 x 7 cm) and a dot with a pencil on the coated side approximately 1.0 cm from the bottom of the strip.

2. Fill a capillary tube (TLC applicator) by placing it in the leaf extract (it will fill by capillary action). Keep your finger off the end of the capillary tube. Apply the extract to the center of the dot on the TLC plate by quickly touching the end of the TLC applicator to the plate. Allow to dry (you can gently blow on the strip). Repeat several times to make a concentrated dot of extract. Be sure to let dry between applications.

3. Carefully place the TLC plate in the TLC chamber (75% petroleum ether, 10% ethyl acetate, 10% acetone, 5% methanol).. The TLC plate should sit on the bottom of the chamber and contact the solvent (the solvent surface must be below the extract dots). Put the lid on the TLC chamber.

4. Allow the TLC plate to develop (separation of pigments) for approximately 10 minutes. As the solvent moves up the TLC plate, you should see the different colored pigments separating.

5. Remove the TLC plate from the chamber when the solvent front is approximately 1.0 cm from the top of the TLC plate. With a pencil, mark the level of the solvent front (at the highest level, the solvent moves up the TLC plate) as soon as you remove the strip from the chamber (the solvent evaporates and disappears quickly). Also, measure the pigment distances quickly, as some pigments (especially the beta-carotene) may fade over time.

11. Record the results of the separation on the datasheet.

**Results of TLC**

1. Draw the TLC plate you have observed in expt. Write the distance of the solvent front from the baseline (from the dot). Write the color of each pigment and the distance from the dot. Label each band as to its pigment. (Beta-carotenes are the most non-polar pigments (highest $R_f$), and their band will be yellow. Chlorophyll *a* & *b* will have lower $R_f$ values). Use these values to calculate $R_f$.

3. The rate at which a pigment moves up the plate is reported as an $R_f$ value, defined as the ratio of the distance moved by the spot to the distance moved by the solvent. Determine the Rf values for each pigment you observe using the formula below.

$$\mathbf{R_f} = \frac{\textbf{Distance moved by solute (pigment)}}{\textbf{Distance moved by solvent}}$$

4. Place your developed TLC plate under the UV lamp (coated side up). Describe what you see. Explain.

## Experiment 3: Separation of Mixture of Organic Compounds by Column Chromatography

First, perform the TLC for the given mixture using the TLC mentioned above and identify a suitable solvent system and the number of compounds in the mixture. Next, do the following:

**Column Preparation:**

1. Assemble a clean, **dry** microscale column.
2. Add a cotton plug at the bottom of the column (just above the stopcock).
3. Ensure the stopcock is closed and add ~5 mL hexane to the column.
4. Prepare a slurry by mixing silica gel (60-120 mesh) with 5-10 mL hexane.
5. Open the stopcock, swirl the slurry (it will settle quickly), then *immediately* pour the mixture into the hexane in the column through a funnel. The silica will gently settle into the column (and the drip rate will slow). Gently tap the column to encourage uniform packing. The collected solvent may be recycled until the slurry has been added to the column.
6. Draw the excess hexane down to the top of the silica gel level and close the stopcock. *Never let the solvent level drop below the top of the adsorbent after the column has been prepared,* or channels may develop.

**Mixture Separation:**

1. Add a mixture of the compounds directly or slurry (dissolve the compound in 2 mL DCM, add 2-3 g of silica gel, and remove the solvent by rotary evaporator) to the column.
2. Add anhydrous sodium sulfate or sand to the top of the column to form a 0.5 cm layer. This is to protect the fine silica as the sample is applied. ***Do not allow the solvent level to drop below the sand or sodium sulfate layer at any time!***
3. Start the elution with the appropriate solvent/solvent combination. You can choose a suitable solvent/solvent combination by performing a TLC experiment. The first band to move down the column is **compound 1**, which should be orange. Start collecting the first portion of the eluent in clean, labeled 10 mL test tubes. Continue adding solvent until the orange band has wholly moved through the column and the drops from the stopcock are colorless.
4. Change the eluting solvent/solvent combination to elute **compound 2**, which should be pale yellow. Collect the second portion of the eluent in a separate clean setup in labeled 10 mL test tubes.
5. Next, pool the first and second portions you collected separately in two separate dry and clean round bottom flasks. Remove the solvent from a rotary evaporator and vacuum dry to get pure compounds.
6. Calculate yields and measure the melting points.

**In your conclusion, discuss the effectiveness of the separation/purification by the column:** Was the separation successful? Were the isolated compounds pure? What proof do you have? *(Hint: use the **visual** results of your TLC plate.*)

**Reference:** A. Johnston *et al. Journal of Chemical Education*, **2013**, *vol 90,* 796.

**Experiment 4: Condensation Synthesis, Characterization, and Photoisomerization of α-Cyanostilbenes**

**Aim:**

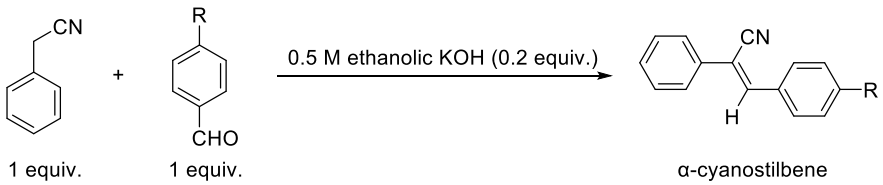
Synthesis of α-cyanostilbenes under an open atmosphere at room temperature via Knoevenagel condensation reaction, characterizations, and study on their photoisomerization properties.

**Theory:**

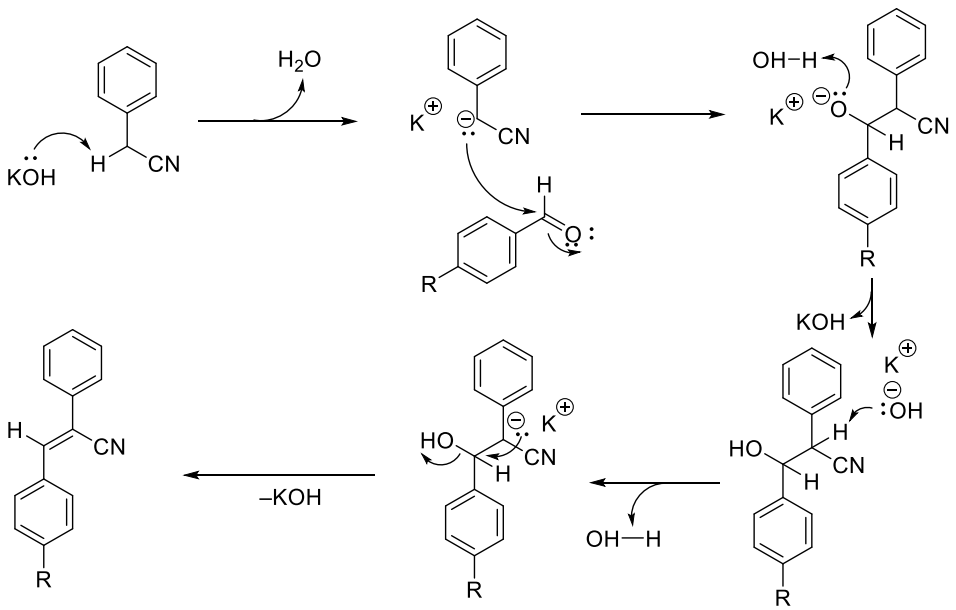
The Knoevenagel condensation is a carbon-carbon bond-forming reaction between a compound containing an active methylene group and an aldehyde or ketone. The product is an α,β-unsaturated compound. This reaction is named after Emil Knoevenagel, a German chemist who first described it in the late 19th century. The reaction is valuable for introducing conjugated double bonds into organic molecules. It has broad utility in various fields, including the pharmaceutical industry, dye formation, natural product synthesis, material chemistry, photochemistry, and polymers.

The reaction typically proceeds in the presence of a base that deprotonates the active methylene compound. The resultant stabilized carbanion attacks the carbonyl carbon of the electrophile to form the carbon-carbon bond. Subsequent water elimination creates unsaturation. In this experiment, phenylacetonitrile is used as the active methylene compound. The deprotonation generates a cyano-stabilized carbanion. Nucleophilic attack and water elimination provided the α-cyanostilbene product.

**Reaction scheme:**




**Mechanism of Knoevenagel condensation:**



**Photoisomerization of *α*-Cyanostilbene:**

Irradiation under UV or visible light, α-cyanostilbene undergoes photoisomerization. Most commonly, *trans*-to-*cis* stereoisomerization takes place around the central olefin bond. The photoisomerization rate depends upon substituents (electronic factors) on olefin, the wavelength of the incident light, light intensity, and some other parameters. The capacity of a chemical compound to absorb light is measured by the molar extinction coefficient or absorptivity (ε) of the compound. The absorptivity is obtained experimentally by calculating the slope from the absorbance vs concentration graph utilizing Beer's-Lambert law.

$A = \varepsilon lc$

Where,

A = absorbance of light

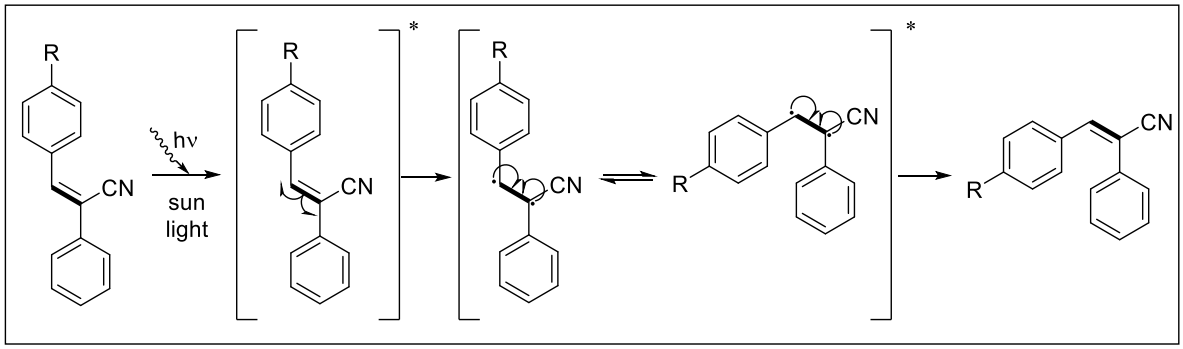
ε = Molar extinction coefficient ($M^{-1}cm^{-1}$)

l = length of the path of light that passes through solution (cm)

c = concentration of solution (M)

*Trans*-α-cyanostilbene goes into the excited state after irradiation. π bond cleaves homolytically in an excited state and generates radical, leaving the σ-bond unaffected. The rotation along this σ-bond gives the kinetically favorable intermediate. Radical recombination quenches the excitation energy to give *cis*-product.

**Mechanism of photoisomerization:**




Sunlight is used as the light source to irradiate α-cyanostilbenes. The sunlight mainly comprises radiations from near-IR (> 700 nm) (50%), visible (400 – 700 nm) (40%) and UV (200-400 nm) (10%) regions. Approximately 95% of UVA (315 – 400 nm) and 5% of UVB (280-315 nm) are covered by sunlight in the UV region.

**Usage of FTIR:**

The functional group present in the organic molecule can be confirmed by FTIR analysis. It mapped the energy required to stretch the bonded atoms. Bond strength and reduced atomic mass determine functional groups' vibration energy or frequency. Hooke's law gives this approximate vibration frequency.

$$\bar{v} = \frac{1}{2\pi C}\sqrt{\frac{K}{\mu}}$$

Where,

$\bar{v}$ = vibration frequency of bond ($cm^{-1}$)

C = speed of light ($2.998 \times 10^{10}$ cm $s^{-1}$)

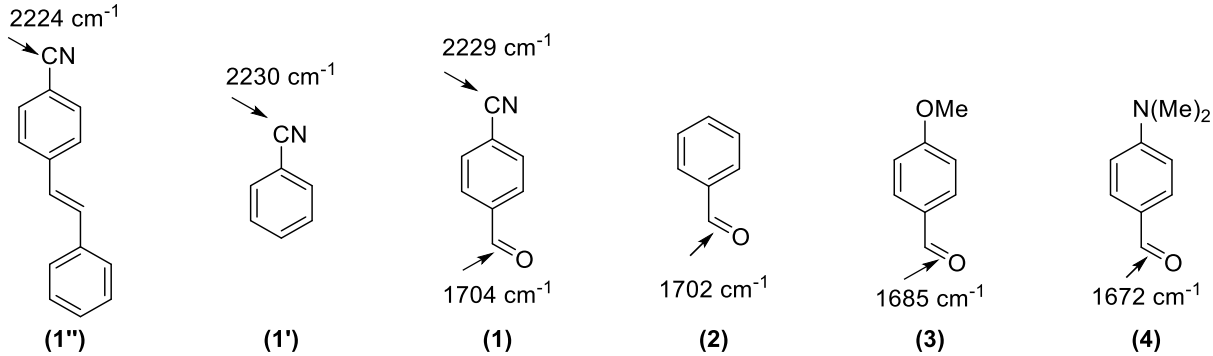
K = force constant of the bond

$\mu$ = reduced mass ($m_1 \times m_2/(m_1 + m_2)$)

$m_1$ and $m_2$ are the mass of two atoms

As bond strength/bond multiplicity/bond order increases, stretching frequency increases. Similarly, decrement in bond strength decreases the stretching frequencies of the bond. The increment/decrease of bond strength depends on resonance, conjugation, inductive effect, mesomeric effect, hydrogen bonding, etc.

**Stretching frequencies of starting materials and some relatable compounds**




**Materials required:-**

**Apparatus:**

50 mL test tube, medium magnetic bar, magnetic stirrer, clamp, glass rod, test tube holder, filter paper, ice, suction pump, Buchner funnel, Buchner flask, water bath, variable volume pipette (100 µL), GC vial, Eppendorf, pre-coated TLC plates (silica gel 60 F254 0.25 mm), TLC chamber, standard volume flask (10 mL, 5 mL), cuvette.

**Chemicals:**

Ethanol, potassium hydroxide, benzaldehyde, 4-methoxybenzaldehyde, 4-cyanobenzaldehyde,

*N*,*N*-dimethyl-4-aminobenzaldehyde, phenylacetonitrile, hexane, ethyl acetate, distilled water, acetonitrile.

**Procedure:**

1. Take a 50 mL test tube and put the medium-sized magnetic bar inside it.
2. Add aldehyde (2.0 mmol) and phenylacetonitrile (2.0 mmol) in the taken test tube carefully.

**Note: Do not contaminate the stock chemicals with each other**

3. Clamp this chemical-containing test tube over the magnetic stirrer and stir it for two minutes to mix the two chemicals homogeneously.

4. After two minutes, add 0.8 mL ethanolic KOH (0.5 M freshly prepared stock solution) into the test tube and let it stir. Note the time of commencement of precipitate formation. Usually, it takes 2-20 minutes for the given reactions in this experiment.

**Note: Observe the product formation once the precipitate starts forming**

5. After 10 minutes, dissolve a pinch of solid crude product or a drop of liquid product in ethyl acetate in an Eppendorf. Monitor the reaction progress via TLC (Thin Layer Chromatography) using the ethyl acetate : hexane (10 : 90) eluent. Measure the $R_f$ value of the starting materials and product.
6. After the reaction (15-20 minutes), add 1-5 mL ethanol into the test tube and heat it in a hot-water bath to dissolve the precipitate to form a clear solution.

**Note: Please check the solubility of products in ethanol. Add slowly and in batches while dissolving the precipitate.**

7. Allow the resulting solution to attain room temperature. As the temperature starts decreasing, crystallization of the product takes place.

**Note: Cool down the test tube in an ice bath if crystallization does not start. Within half an hour, crystallization will be complete.**

8. Filter the resulting crystals and wash them with plenty of water first, then 1-5 mL cold ethanol using normal filter paper and Buchner funnel.

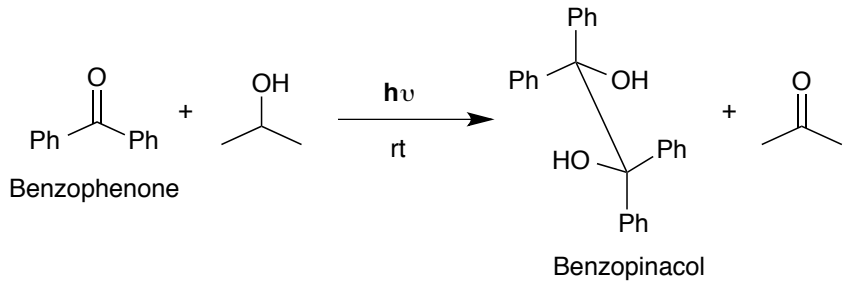
**Note: Crystals of B have a higher solubility in ethanol. Carefully wash it with a minimum amount of cold ethanol.**

9. Dry the filtered crystals under air and press them under two ordinary filter papers.
10. Once it dries, note the color of the product, measure the weight, and calculate the yield.
11. Record the melting point of the dried product and match it with the reported value.
12. Record the FT-IR spectra of the product. After getting the recorded spectra, mention the stretching frequency of the olefinic CN functional group while processing the spectra.
13. Prepare 5 mL 5-25 μM solutions of the product in acetonitrile solvent. Get the UV-Vis spectrum recorded and note down the $\lambda_{max}$ value. Using these spectra, plot the absorbance vs concentration graph. Then, calculate the slope to determine the molar extinction coefficient using Beer's - Lambert Law.

**Note: Instantly measure the UV-Vis spectra after preparing the diluted solution. Otherwise, *trans*-to-*cis* isomerization could start under the ambient light inside the lab or any light source. If this happens, $\lambda_{max}$ will shift towards a lower wavelength.**

14. Freshly prepare 25 mM solution of the product in acetonitrile and take 1.0 mL solution in GC-vial. Get the sample analyzed via GC-MS. After analysis, the same GC-vial was put under sunlight for 0.5 h. Again, get it analyzed via GC-MS. Match the spectra before and after irradiation and note the retention time of the observed peaks. Also, measure the *trans*:*cis* ratio of the product quantitatively.
15. Tabulate and compare the FT-IR, UV-Vis, and GC-MS data with other groups. Note the trends in value changes and discuss the reasons behind them.

**Experiment 5: Preparation of Benzopinacol by Photolysis**



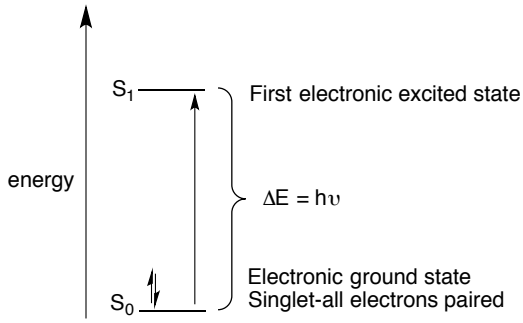

The chemistry of reactions that are initiated by light is *photochemistry*. The energy in light is proportional to the light's frequency. Since the frequency and wavelength of light are related, the energy of light may be expressed as either a function of frequency or wavelength.

$$E = h\nu = hc/\lambda$$

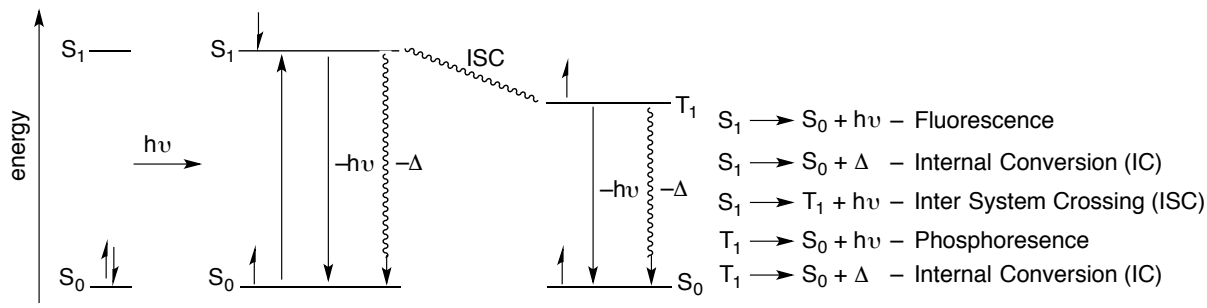
$E$ = energy of light, $h$ = Planck's constant, $\nu$ = frequency, $\lambda$ = wavelength and c = speed of light

Many molecules can absorb light. A photon of light is absorbed by a chemical only if the energy of the photon corresponds exactly to the difference in electronic levels (i.e., orbitals) in a molecule. If all the electrons are in the lowest possible energy levels, then the molecule is in its ground state ($S_o$). Notice that the electrons are spin-paired in the ground state.

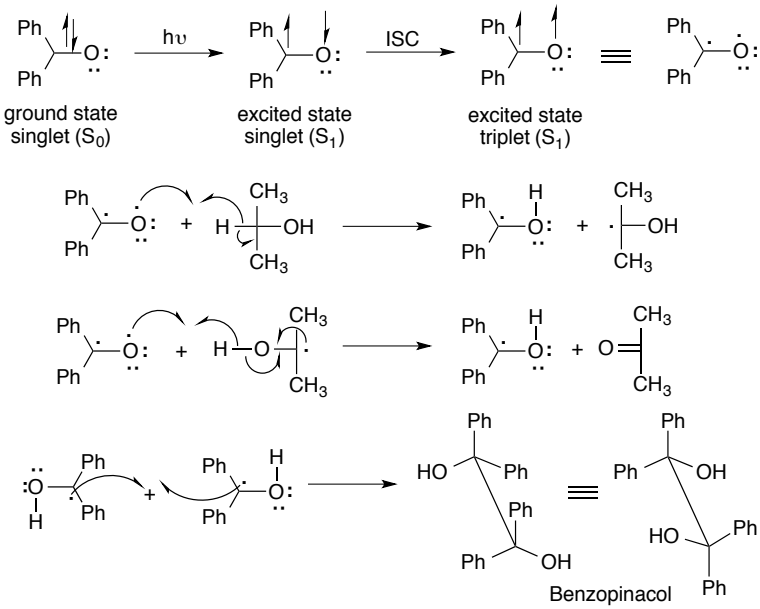
The electrons in the highest energy molecular orbital are most likely to be excited by a photon. Absorption of light of the appropriate wavelength (355 nm for benzophenone) promotes one of the electrons in the Highest Occupied Molecular Orbital (HOMO) to a higher energy orbital (often the Lowest Unoccupied Molecular Orbital - LUMO). Once an electron jumps into a higher energy orbital, then the molecule is in an excited state. Initially, the spin of the electron is unaltered. That means it will still have a spin opposite to the one-electron remaining in the lower energy orbital. (The electrons are still spin-paired) As long as the spins of the electrons remain paired, the molecule is in a *singlet state*. The first excited singlet spin state is denoted $S_1$.




**Jablonski Diagram:**



From this point, several things could happen. The electron in the excited state could emit energy as light and return to the electronic ground state, a process known as *fluorescence*. An alternative is for an *intersystem crossing* (ISC) to occur. ISC occurs when the electron in the excited state *changes the direction of its spin*. The result is that the electrons are no longer spin-paired, resulting in a *triplet state,* $T_1$. Intersystem crossing is faster than fluorescence for benzophenone, so fluorescence is not observed. Once an intersystem crossing occurs, the molecule will either react with another chemical or return to its ground state electronic configuration in a process known as *phosphorescence*. Phosphorescence is relatively slow, so the molecule has time to react. (Note that the excited electron must undergo another spin-flip during phosphorescence, and energy is released in the form of light during phosphorescence)




**Mechanism**: In this experiment, benzophenone is allowed to dissolve in 2-propanol, and the flask is placed in the sun. 2-Propanol acts as both a reagent and the solvent in this reaction. The benzophenone absorbs ultraviolet light and goes into an excited *singlet* state followed by intersystem crossing to produce a *triplet* diradical (a molecule with two unpaired electrons). The benzophenone diradical then abstracts a hydrogen atom from the solvent and dimerizes, yielding benzopinacol and acetone.

**Experimental Procedure:**

**Week 1:** In a 10 mL test tube, dissolve 1 g benzophenone (mp 48 °C) in approximately 7 mL of 2-propanol (bp 82 °C), gently warming the test tube to aid dissolution. After all the solid is dissolved, add 2-3 mL 2-propanol and one drop of glacial acetic acid (to prevent basic cleavage of the desired product). Using a small label, label the test tube with your name, insert a cork, and wrap parafilm around the mouth of the test tube. Place the flask in a safe, sunny location. The first product (Benzopinacol; mp 189 °C) should crystallize in about five hours, and the reaction should be ~ 95% complete in four days.

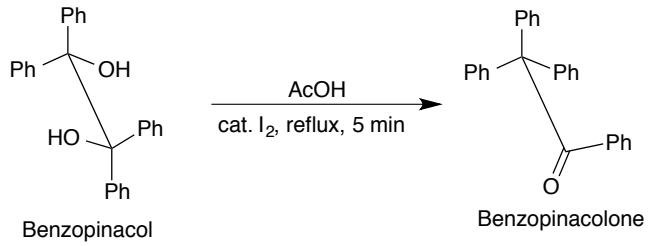
**Week 2:** Chill the test tube in an ice bath and then collect the product by vacuum filtration. The product should be pure at this point. Allow the crystals to dry overnight, get an accurate mass and melting point, and submit your product to your TA. Be sure to calculate a percent yield for your product and **SAVE**

**THE BENZOPINACOL FOR THE PINACOLONE REARRANGEMENT REACTION**!

**Waste**: Any 2-propanol waste, including the filtrate, may be flushed down the drain. Any recovered starting material should be placed in the solid waste jar.

**Reference**: Williamson, *Macroscale and Microscale Organic Experiments, 3rd Ed.*, Houghton Mifflin Company, **1999**.

**Experiment 6: Preparation of Benzopinacolone by Pinacol Rearrangement**




**Experimental Procedure:**

In a 25 mL round-bottomed flask, place 0.5 g of benzopinacol, 5 mL of acetic acid, and 1 tiny iodine crystal. Equip the flask with a reflux condenser and heat to boiling for 1-2 min until the crystals dissolve. Once dissolved, reflux for 5 min. Upon cooling, benzopinacolone separates as a thick paste. Add a small amount of ethanol, mix thoroughly, and isolate your product by vacuum filtration using a Büchner or Hirsch funnel. Rinse the flask and crystals with additional ethanol to remove any remaining iodine color. Allow your product to dry in your drawer. Determine the yield and m.p. of the crude product and compare with the literature value.

**Cleaning up:**

Dilute the filtrate with water. Neutralize with sodium bicarbonate (baking soda) and flush down the drain.

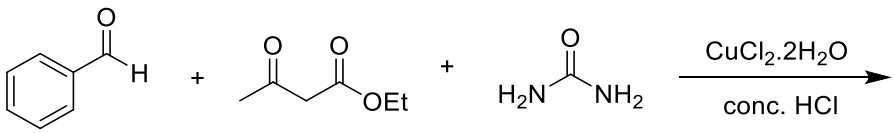
**Reference:** Williamson, *Macroscale and Microscale Organic Experiments, 3rd Ed.*, Houghton Mifflin Company, **1999**.

**Experiment 7: Grindstone Chemistry: Synthesis of Heterocyclic Compounds by Multicomponent Reaction Using Biginelli Reaction**

Multicomponent reactions and heterocyclic chemistry are important concepts of organic synthesis, especially in the pharmaceutical industry. Biginelli reaction is a one-pot condensation of a 1,3-dicarbonyl compound, aryl aldehyde, and urea in the presence of a Lewis acid to form a dihydropyrimidine. Dihydropyrimidine derivatives show many medicinal properties, such as calcium channel blocking, antiviral, and antibacterial activities.

Safe, practical, and environmentally friendly "Green Chemistry" techniques are required to reduce environmental pollution. Accomplishing the reaction by grinding the starting material without using solvents (grindstone chemistry) is one of the best green chemistry techniques.

**Reaction Scheme:**




(Ethyl acetoacetate: mol. wt. 130, density: 1.02 g/cm³; Urea: mol. wt. 60, $CuCl_2.2H_2O$: mol. wt. 170.48)

**Experimental Procedure:**

Grid together a mixture of an aromatic aldehyde (5 mmol), ethyl acetoacetate (5 mmol), urea (10 mmol), $CuCl_2 \cdot 2H_2O$ (2.5 mmol), and 3-4 drops of conc. HCl for 5 minutes. Keep grinding occasionally for another hour. Then, wash the solid mass obtained with cold water and purify by crystallization from ethanol. Record melting point and IR spectrum.

***Reference:*** *Indian Journal of Chemistry,* **2008***, 47B,* p434.

**Experiment 8: Diels-Alder Reaction of Anthracene and Maleic Anhydride**

**Aim and Objectives:** In this lab, we will learn that the 4π electrons of the diene (anthracene) will react with the 2π electrons of the dienophile (maleic anhydride) to give the polycyclic Diels-Alder product.

**Apparatus Requirement:**

Round Bottom (RB) flask; Reflux condenser; Celite funnel; Measuring cylinder; Filter paper

**Chemicals Requirement:**

1. Anthracene
2. Maleic Anhydride
3. *o*-Xylene

**Molecular Wt. and Physical Details of Reagents:**

The molecular formula of Anthracene =$C_{14}H_{10}$

Molecular weight of Anthracene = 178.2340 g/mole

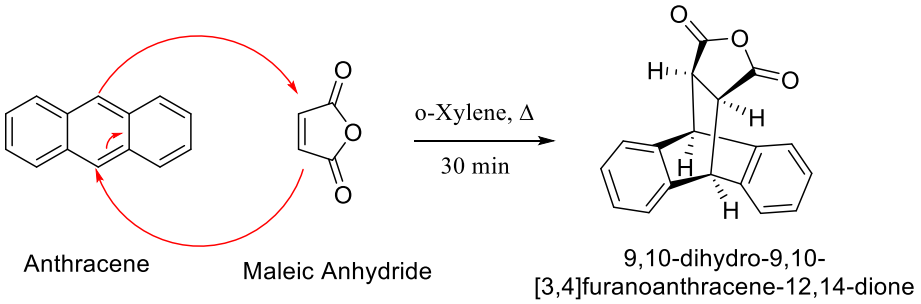
The molecular formula of Maleic Anhydride = $C_4H_2O_3$

The molecular weight of Maleic Anhydride = 98.0570 g/mole

**Experimental Procedure:**

1. Anthracene (1.0 g) was taken in a cleaned and dried round bottom (RB) flask.
2. Maleic anhydride (0.5 g) was taken in the same round bottom flask.
3. o-Xylene (10 mL) was added to the reaction mixture, and the contents of the flask were mixed thoroughly.
4. Then, the RB was warmed with a heating-bath previously maintained at 140 °C for about 25 min with constant stirring.
5. After 25 min, the heating bath was removed, and the reaction mixture was allowed to cool to room temperature.
6. After a few minutes, the RB was transferred to an ice bath for further cooling and left for 10 minutes.
7. After 10 minutes, a white precipitate was observed, which was filtered through a celite funnel using a vacuum.
8. The precipitate was washed with ice-cold xylene and drained well to give a crude Diels-Alder product.
9. The weight of the crude product was taken.
10. The melting point of the product was checked.
11. Further characterization was done by IR spectra, $^{1}H$ NMR spectra, and $^{13}C$ NMR spectra.

**Reaction Mechanism:**




**Results and Observation:**

Yield: 1.43 g % of Yield: 96.95% MP: 257- 259 °C

TLC System: $R_f$ = 0.44 (20% EtOAc in *n*-hexane)

**Compound Characterization:**

**IR Frequencies:** 1839, 1757, 1441, 1212, 1053 $cm^{-1}$

**$^{1}$H NMR:** (400 MHz, DMSO-d6) δ 7.47 (dd, J = 5.4, 3.2 Hz, 2H), 7.36 – 7.29 (m, 2H), 7.18 (td, J = 5.9, 3.2 Hz, 4H), 4.86 (s, 2H), 3.65 (s, 2H)

**$^{13}$C NMR:** (126 MHz, DMSO-d6) δ 176.75, 146.34, 144.32, 132.29, 131.76, 130.08, 129.69, 53.14, 49.56.

**Conclusion:**

In summary, we have done the Diels-Alder reaction of anthracene and maleic anhydride. The chemical yield was 96.95%. Further, the compound was characterised by IR spectra, NMR spectra and melting point was also checked.

**Experiment 9: Chemical Synthesis of Analgesic and Antipyretic Drug, Paracetamol**

**Aim and Objectives:** Chemical Synthesis of Analgesic and Antipyretic Drug, Paracetamol from *p*-aminophenol and acetic anhydride.

**Apparatus Requirement:**

Round Bottom (RB) flask; Celite funnel; measuring cylinder; Filter paper

**Chemicals Requirement:**

1. *p*-Aminophenol
2. Acetic anhydride
3. Concentrated sulphuric acid
4. Charcoal
5. Ethanol

**Molecular Wt. and Physical Details of Reagents:**

The molecular formula of *p*-aminophenol = $C_6H_7NO$

Molecular weight of *p*-aminophenol = 109.13 g/mole

The molecular formula of Acetic Anhydride = $C_4H_6O_3$

The molecular weight of Acetic Anhydride = 102.09 g/mole

**Experimental Procedure:**

1. 1 g of *p*-aminophenol was weighed and taken in a cleaned and dried round bottom flask.
2. 1 mL of acetic anhydride was measured accurately in a glass measuring cylinder.
3. 1 drop of concentrated sulphuric acid was added to the reaction mixture carefully, and the contents of the flask were mixed thoroughly.
4. Next, the RB flask was warmed with a water-bath previously maintained at 60 °C for about 20 min with constant stirring.
5. Then, the water bath was removed, and the reaction mixture was allowed to cool to room temperature.
6. 20 mL of cold water (with a few chips of crushed ice) was added and stirred vigorously.
7. The white precipitate was observed and filtered through a celite funnel using suction.
8. The precipitate was washed with plenty of cold water, drained well, and dried in an electric oven at 100 °C.
9. The weight of the crude paracetamol was taken.

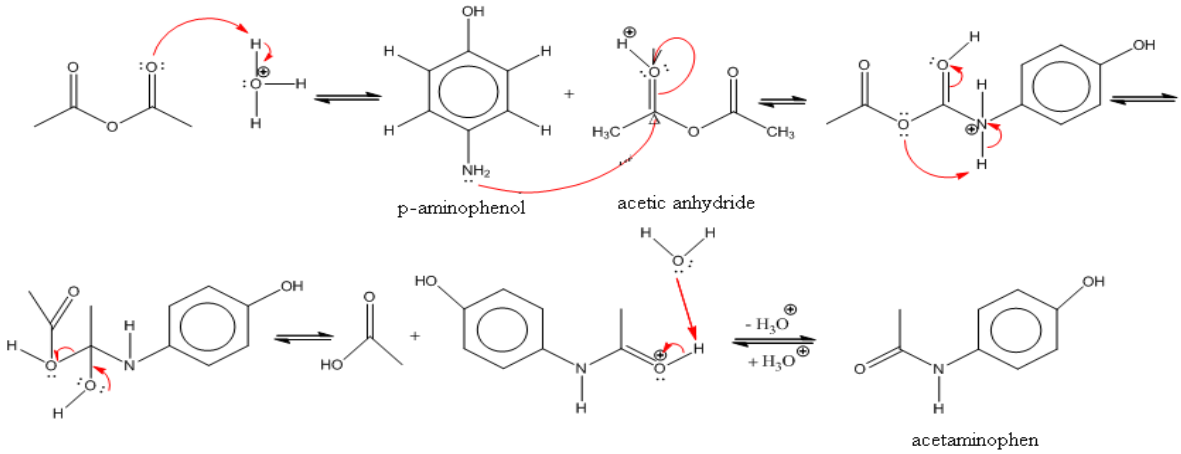
**Recrystallization Procedure:**

1. The crude paracetamol was taken in 70% (v/v) ethanol and was warmed to 60°C until fully dissolved.

2. Then 200 mg of powdered animal charcoal (decolorizing carbon) was added.

3. The mixture was then filtered, the filtrate was concentrated over a water bath and allowed to cool, and large monoclinic crystals were separated.

**Reaction Mechanism:**




**Results & Observation:**

Yield: 1.2 g % of Yield: 85% MP: 172-175 °C

TLC System: $R_f$ = 0.38 (50% EtOAc in *n*-hexane)

**Compound Characterization:**

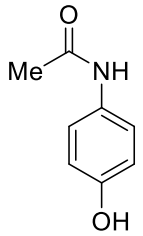
**IR Frequencies**: $\nu$ = 3280, 3068, 1629, 1589, 1542, 1486, 1417, 1352, 1209, 825, 673 $cm^{-1}$.

**$^1$H NMR**: (**400 MHz**, DMSO-D6) δ 9.60 (s, 1H), 9.09 (s, 1H), 7.29 (d, *J* = 9.1 Hz, 2H), 6.63 (d, *J* = 9.1 Hz, 2H), 1.94 (s, 3H).

**$^{13}$C NMR**: (**100 MHz**, DMSO-D6) δ 168.0, 153.6, 131.5, 121.3, 115.5, 24.3.

**Conclusion:**

In summary, we have completed the chemical synthesis of analgesic and antipyretic drug paracetamol from a reaction of *p*-aminophenol and acetic anhydride. The chemical yield was 85%. Further, this compound was recrystallized using ethanol and charcoal.



paracetamol (**1**)

| **other names:** |
|---|
| acetaminophen (**1a**) |
| *N*-acetyl-*p*-aminophenol (**1b**) |
| *p*-hydroxy acetanilide (**1c**) |

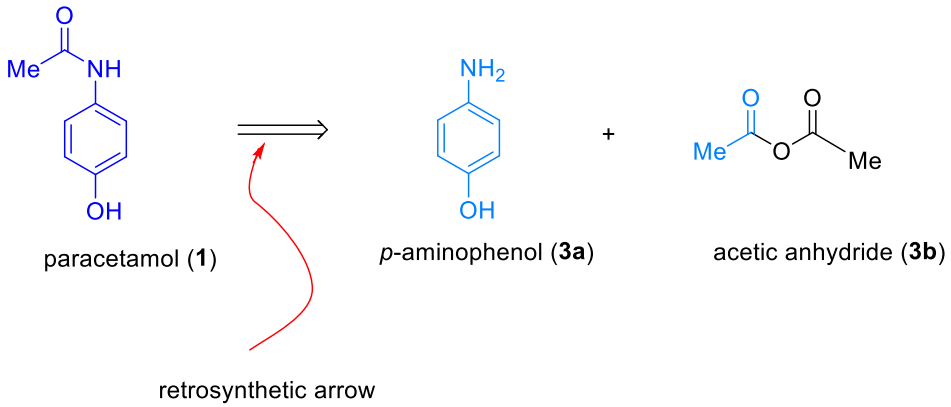
| **Trade names:** |
|---|
| Calpol (**2a**) |
| Panadol (**2b**) |
| Tylenol (**2c**) |
| Excedrin (**2d**) |

Calpol (2a), Panadol (2b), Tylenol (2c), and Excedrin (2d) are brand of drugs advertised for reducing pain, reducing fever, and relieving the symptoms of allergies, cold, cough, headache, and influenza.

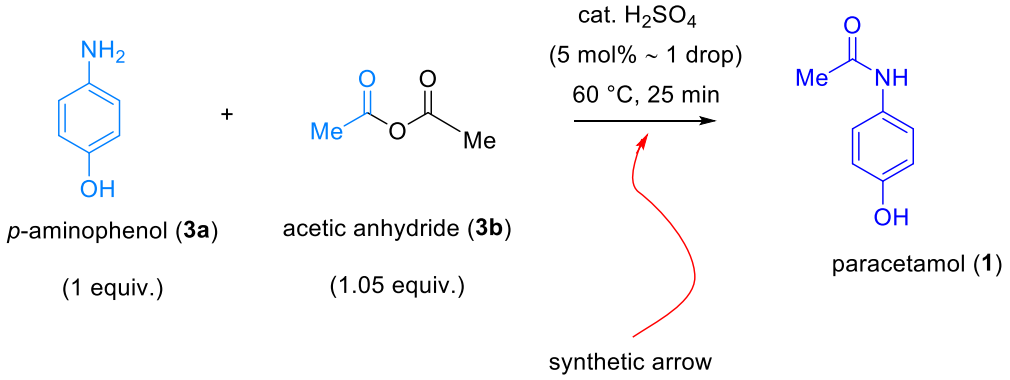
The active ingredient of its original flagship product is paracetamol, an analgesic and antipyretic.

The brand name Tylenol is derived from a chemical name for the compound, *N*-acetyl-*p*-aminophenol (APAP).

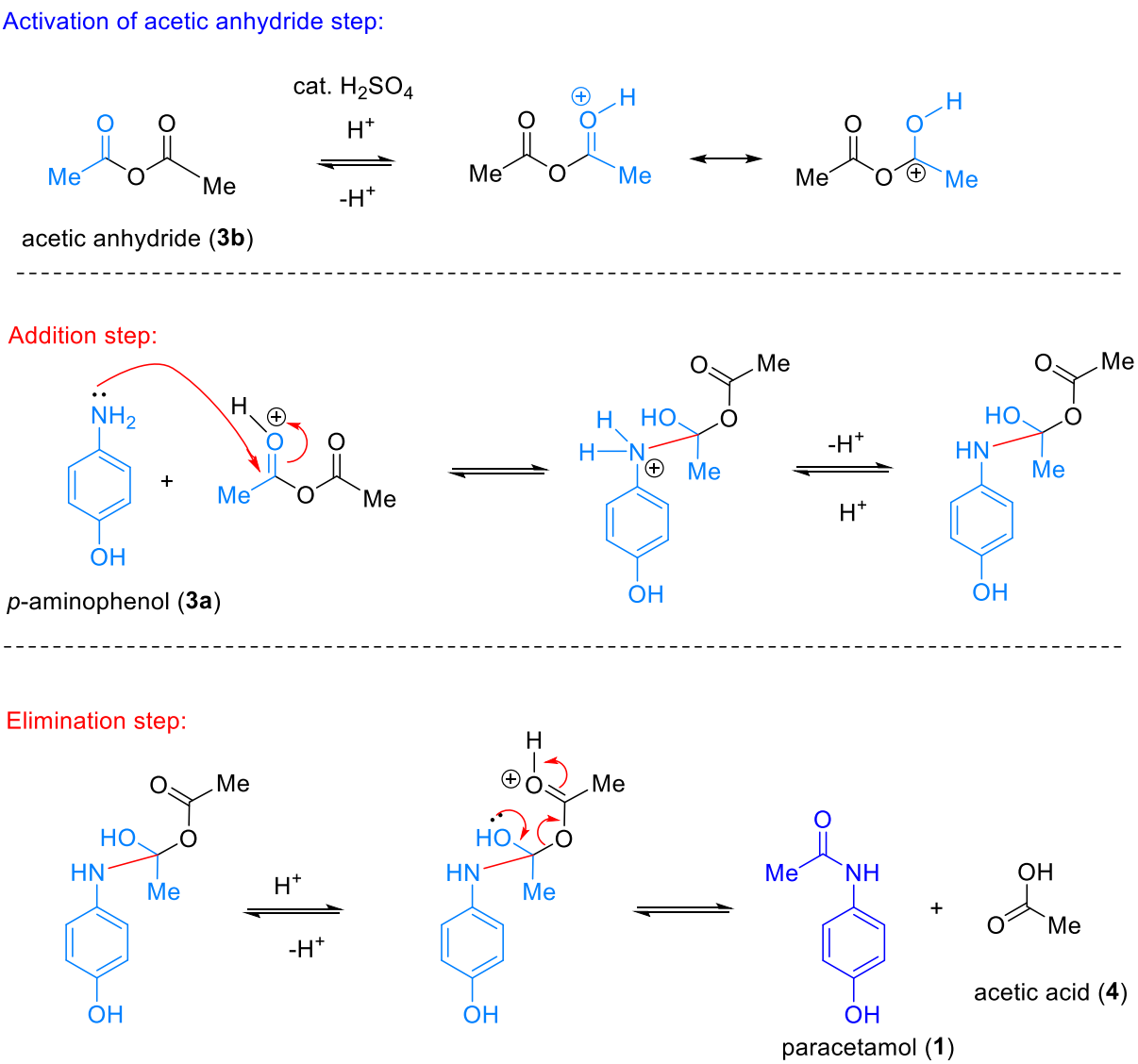
**Figure 1:** Paracetamol.




**Scheme 1:** Retrosynthetic analysis of Paracetamol.



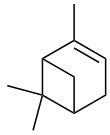

**Scheme 2:** Reaction of *p*-aminophenol and acetic anhydride.




**Scheme 3:** Reaction mechanism of paracetamol synthesis.

**Experiment 10: Iodination of Cyclohexene and α-Pinene**

In *All Creatures Great and Small*, James Herriot describes how veterinarians often disinfected wounds in large animals. The veterinarian would pack the wound with iodine, splash some turpentine, and stand back. A violent reaction forced gaseous iodine into the wound amid a cloud of purple vapor. This was accompanied by an equally violent reaction from the animal. The major component of turpentine is α-pinene, a bridged bicyclic compound containing both a 4-membered ring and a 6-membered ring. Although alkenes readily undergo addition reactions with chlorine and bromine, most are unreactive with iodine. If α-pinene is responsible for the reaction of turpentine with iodine, and it must possess some structural feature that makes it more reactive than other alkenes. You will investigate the effect of adding α-pinene to iodine and compare it to the reaction of cyclohexene and iodine.



α-Pinene

Cyclohexene

**Experimental Procedure:**

***CAUTION:*** **α-Pinene is flammable and irritates the skin and eyes. The vapors are potentially harmful. Minimize contact and do not breathe vapors. The reaction with iodine may be violent, conducted under a fume hood. Wear safety goggles and adequate protective clothing.**

Into two 15 cm test tubes, weigh 0.5 g iodine into each *(TOP LOADING BALANCE ONLY! Do not use the analytical balances.)* Under the hood, add 1.0 mL of cyclohexene to the first test tube in one portion. Then, add 1.0 mL of α-pinene to the second test tube in one portion, step back quickly, and observe the reactions.

Write a mechanism that explains the exothermicity of the α-pinene reaction, assuming a carbocation as the initial intermediate. Write an equation showing two viable products (neither is vicinal diiodide). Tell how your mechanism explains your experimental observations.

Why is the addition of α-pinene to iodine so exothermic? Would you expect the bromination of α-pinene to be more or less exothermic? Explain the reaction mechanism.

**Reference:** John W. Lehman, *Multiscale Operational Organic Chemistry*, Prentice Hall, **2002**, p 480.